Manip — cheat sheet

Keyboard-driven algebraic manipulation. Lowercase letters, digits and symbols apply operations directly to the formula (the same keys work in the Enter menu); uppercase (Shift) are commands.

NAVIGATION AND STEPS

← → ↑ ↓	navigate
Ctrl ← →	move / take to the other side
Ctrl ↑ ↓	leave / enter group
Shift ← →	select neighbors
Space	select
Esc	clear selection / leave
Enter	operations
Tab	workspaces
PgUp PgDn Home End	go through the steps
Ctrl Z	undo
Ctrl Shift Z	redo

COMMANDS (SHIFT)

C	color
E	edit
Y	copy the expression (text)
L	copy LaTeX
P	present
K	keyboard
O	step note
R	record video
]	larger formula
[	smaller formula
\	normal size
?	help: shortcuts and input syntax

ARITHMETIC

c	Calculate ↔ factor the number (12 ↔ $2^2 \cdot 3$)	$x = 7 - 3 \;\to\; x = 4$
l	Combine like terms	$x = 2a + 3a \;\to\; x = 5a$
x	Cancel	$x + 3 - 3 = 5 \;\to\; x = 5$
0	Identity and zero: x + 0, 1·x → x; 0·x → 0; $x^0 \to 1$; $1^k \to 1$	$x + 0 = y \;\to\; x = y$
r	Sign rule: −(−a), (−a)b ↔ −ab	$x = a - (-b) \;\to\; x = a + b$
-	Flip signs: A(B + C) → −A(−B − C), −(B + C) ↔ (−B − C), −∫ f dx ↔ ∫ −f dx, −Σ f ↔ Σ −f	$2(x - 1) = 4 \;\to\; -2(-x + 1) = 4$

STRUCTURE

d	Distribute	$2(x+1)=4 \;\to\; 2x+2\cdot 1=4$
e	Factor out	$1-a-b+ab=0 \;\to\; 1-a-b(1-a)=0$ with selection
(	Wrap in parentheses	$a+b+c=0 \;\to\; a+(b+c)=0$ with selection
p	Remove parentheses	$a+(b+c)=0 \;\to\; a+b+c=0$
g	Group selection	$a+b+c=0 \;\to\; a+(b+c)=0$ with selection
q	Add and subtract…	$x^2+6x=7 \;\to\; x^2+6x+9-9=7$ typing “9”
s	Substitute with…	$x+y=10 \;\to\; x+2x=10$ typing “2x”

FRACTIONS

j	Fractions: a/c + b/c ↔ (a + b)/c, (a/b) (c/d) ↔ ac/(bd)	$\frac{x}{2}+\frac{y}{2}=1 \;\to\; \frac{x+y}{2}=1$
*	Multiply top and bottom by…	$x=\frac{1}{2} \;\to\; x=\frac{1\cdot 3}{2\cdot 3}$ typing “3”

POWERS

^	Powers and roots: aa ↔ a^2, $x^a x^b$ ↔ x^(a+b), $a^n b^n$ ↔ $(ab)^n$, √a√b ↔ √(ab), √x ↔ x^(1/2)	$x^2x^3=1 \;\to\; x^5=1$
b	Binomial / multinomial ↔ special product; $(a+b)^n$ ↔ Σ C(n, k) a^{n-k} b^k	$(a+b)^3=0 \;\to\; a^3+3a^2b+3ab^2+b^3=0$

FUNCTIONS

f	Log and exp: log(ab) ↔ log a + log b, log a^n ↔ n log a, ln e^x → x, change of base	$y=\log(ab) \;\to\; y=\log a+\log b$
f	Trigonometry: known values (0, π/6, π/4, π/3, π/2…), $\sin^2+\cos^2=1$, tan = sin/cos, double angle, angle sum, parity	$y=\sin(2x) \;\to\; y=2\sin x\cos x$

SUMS AND COMBINATORICS

!	Factorial and binomial: n! ↔ n·(n − 1)!, C(n, k) ↔ n!/(k!(n − k)!), symmetry, Pascal	$y=\binom{n}{k} \;\to\; y=\frac{n!}{k!(n-k)!}$
m	Sums and products: terms, linearity, limits, index, formulas	$S=\sum_{k=1}^{n}(2k+1) \;\to\; S=\sum_{k=1}^{n}2k+\sum_{k=1}^{n}1$

CALCULUS

'	Derivatives: linearity, power, product, quotient, chain, table, definition, f′ ↔ d/dx	$y=\frac{\mathrm{d}}{\mathrm{d}x}(x^2\sin x) \;\to\; y=\left(\frac{\mathrm{d}}{\mathrm{d}x}x^2\right)\sin x+x^2\frac{\mathrm{d}}{\mathrm{d}x}\sin x$
a	Limits: direct substitution (refused on 0/0), limit laws, notable limits, L'Hôpital, c/x^n → 0, change of variable, definition of the derivative	$y=\lim_{x\to 2}(x^2+1) \;\to\; y=2^2+1$
u	Integrals: table (+ C once solved), linearity, $[F]_a^b$ = F(b) − F(a), join intervals	$y=\int_0^1 x^2\,\mathrm{d}x \;\to\; y=\left[\frac{x^3}{3}\right]_{x=0}^{1}$
u	Integration by substitution: u = g(x)	$y=\int 2x\cos(x^2)\,\mathrm{d}x \;\to\; y=\int\cos u\,\mathrm{d}u$ typing “u = x^2”
u	Integration by parts: u = f(x)	$y=\int xe^x\,\mathrm{d}x \;\to\; y=xe^x-\int e^x\frac{\mathrm{d}}{\mathrm{d}x}x\,\mathrm{d}x$ typing “u = x”
u	Definite integral: swap the limits, Riemann sum	$y=\int_0^1 x^2\,\mathrm{d}x \;\to\; y=-\int_1^0 x^2\,\mathrm{d}x$

COMPLEX NUMBERS

z	Conjugate properties	$w=\overline{ab} \;\to\; w=\overline{a}\,\overline{b}$
~	z ↔ $\bar{\bar{z}}$ (double conjugate)	$w=\bar{\bar{z}} \;\to\; w=z$
h	z·$\bar{z}$ ↔ $\lvert z\rvert^2$	$w=z\bar{z} \;\to\; w=\lvert z\rvert^2$
v	z ± $\bar{z}$ ↔ 2Re(z) / 2i·Im(z)	$w=z+\bar{z} \;\to\; w=2\,\mathrm{Re}(z)$
y	Multiply by the conjugate	$w=\frac{1}{3+4\mathrm{i}} \;\to\; w=\frac{1(3-4\mathrm{i})}{(3+4\mathrm{i})(3-4\mathrm{i})}$

EQUATION AND INEQUALITY

=	Apply to both sides...	$2x = 4 \;\rightarrow\; \frac{2x}{2} = \frac{4}{2}$ typing "/ 2"
t	Swap sides	$x < 4 \;\rightarrow\; 4 > x$
i	Estimate (≤, ≥, <, >)...	$y = x^2 + 1 \;\rightarrow\; y \geq 1$ typing ">= 1"

SYSTEMS

+	Add equations (select two)	$\begin{cases} x+y=3 \\ x-y=1 \end{cases} \rightarrow \begin{cases} x+y=3 \\ x-y+x+y=1+3 \end{cases}$ with selection
-	Subtract equations (select two)	$\begin{cases} x+y=3 \\ x-y=1 \end{cases} \rightarrow \begin{cases} x+y=3 \\ x-y-(x+y)=1-3 \end{cases}$ with selection
>	Substitute this variable into the other equations	$\begin{cases} y=2x+1 \\ x+y=7 \end{cases} \rightarrow \begin{cases} y=2x+1 \\ x+(2x+1)=7 \end{cases}$

NAMES

n	Expand the name	
n	Name it...	$2(x+1)=y \;\rightarrow\; 2\boxed{K}=y$ typing "K"
w	Open in workspace	

Generated by Manip itself: the examples are the operations actually applied.